# Advanced Programming Practice · 21CSC203P

Unit 1 · Session 2 · SLO 1 — Solved Worksheet

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## Unit – 1 : Session – 2 : SLO – 1 | Language Theory Concepts

### Activity (Solutions)

• 1) Type checking verifies that operations are applied to compatible types. STATIC typing (Java): types are checked at compile time — int x = 5; x = "hi"; fails to compile. DYNAMIC typing (e.g., Python): types are checked at run time — x = 5; x = "hi" is legal. Java demo: String s = "abc"; int n = s.length(); // OK (length() returns int); int n2 = s; // compile-time type error.

• 2) Variable scope demo: public class Scope { static int g = 10; // class scope public static void main(String[] a){ int x = 5; // method scope if (x > 0) { int y = 2; // block scope System.out.println(x + y + g); } /* y not visible here */ } }

• 3) Binding time vs run time: Binding time is when an association (name→type, variable→value, method→body) is made — e.g., 'int x' binds x to type int at compile time; static methods bind at compile time. Run time is when the program executes — dynamic (polymorphic) method calls bind to the actual method body at run time. Example: Animal a = new Dog(); a.sound(); — the reference type binds at compile time, but which sound() runs is decided at run time.